

Tournament Manager App - Documentation

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1 Introduction

This application is my submission for the final project of the advanced data structures class.

This is a CLI-based application made in C++. It is a simple tournament or contest manager, which holds a leaderboard of players, a queue of upcoming matches sorted by priority, and updates the leaderboard every time a match has been played.

The application supports:

- player management,
- automatic leaderboard updates,
- match scheduling by priority,
- saving and loading the tournament state from files,
- a verbose switch that prints out the step by step operations that are performed behind the scenes.

2 Data structures used

For this project, two data structures were used: the red-black tree, and the binomial heap.

The red-black tree is the structure behind the leaderboard. Each node stores a player's name and score, and, of course, it's color. The tree automatically ballances itself after insertions and deletions, using rotations and re-coloring when necessary.

The Red-Black tree was used for the leaderboard, because it is very efficient for search operations, since the collorings and their rules make it more ballanced, and therefore more efficient, than a standard binary search tree. This structure also guarantees a complexity of $O(\log n)$ for both insertions and deletions.

The match queue is implemented using a custom binomial heap. Each node stores a match's priority and ID.

The heap organizes the matches such that the minimum priority match can always be extracted efficiently.

The binomial heap was used, because it is very efficient, allows merging, and many real-world priority queues are implemented using this structure.

3 How to run

The simplest way to run the application is to open it in C Lion (the IDE in which it was made) and simply build and run it.

The app may also simply be compiled using the cmake scripts provided.

The app offers one command-line argument switch, `-v` or `-verbose`. To use it, open the command line in the directory with the final executable, and run it like this: **FinalProject.exe -v**

If the verbose mode is enabled, the app will print things such as rotations, node colorings, etc when they happen in the backend.

Once the app has been launched, the user is presented with a command line menu. Typing `help` at any point will print again the list of available commands, and `exit` will close the program. The app supports adding new players, scheduling new matches, showing the visual representation of each structure, and saving or loading the tournament state from files.